

Semiconductor English Jargon Quick Reference

Field-specific terms, contrast pairs, and high-pressure sentence frames

Audience: semiconductor process engineers, yield engineers, product engineers, equipment engineers, fab supervisors, quality teams, test and packaging teams, foundry coordinators, supply planners, applications engineers, and technical program managers

Focus: A semiconductor English curriculum for wafer fabrication, lithography, process integration, deposition and etch, metrology, yield learning, cleanroom discipline, equipment uptime, packaging, reliability qualification, foundry communication, and customer pressure.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

Wafer Fabrication Flow and Process Integration

Term	Working meaning
wafer	Thin semiconductor substrate, usually silicon, on which integrated circuits are fabricated.
fab	Semiconductor fabrication facility where wafers are processed through manufacturing steps.
process flow	Ordered sequence of semiconductor manufacturing steps, layers, inspections, holds, and decision points.
node	Technology generation or process family, often associated with feature size, performance, density, and design rules.

Lithography, Reticles, and Critical Dimensions

Term	Working meaning
lithography	Patterning process that transfers circuit features to a wafer using light, masks, and photoresist.
photoresist	Light-sensitive material used in lithography to define patterns on a wafer.
reticle	Photomask used in lithography to project circuit patterns onto a wafer.
critical dimension	A measured feature size on a wafer that must stay within specification for device performance and yield.

Deposition, Etch, CMP, and Process Windows

Term	Working meaning
deposition	Process of adding material layers to a wafer by physical, chemical, epitaxial, or atomic-layer methods.
etch	Process that removes selected material from a wafer using wet chemistry or plasma-based methods.
CMP	Chemical mechanical planarization; a process that smooths wafer surfaces for later manufacturing steps.
process window	Range of process conditions under which results meet specification with acceptable margin.

Metrology, SPC, and Yield Learning

Term	Working meaning
metrology	Measurement discipline used to verify process, dimension, film, defect, and device characteristics.
SPC	Statistical process control; use of control charts and limits to monitor process stability.
yield	Share of wafers, die, units, or lots that meet requirements after manufacturing, test, or qualification.
excursion	Manufacturing event or trend outside expected control limits, specifications, or normal process behavior.

Defect Density and Cleanroom Contamination

Term	Working meaning
defect density	Number or rate of defects on a wafer, die, layer, lot, or process area.
particle	Small contaminant that can create defects, yield loss, reliability risk, or process instability.

Term	Working meaning
cleanroom	Controlled manufacturing environment designed to limit particles, humidity, electrostatic risk, and contamination.
contamination control	Practices used to prevent particles, residues, metals, organics, moisture, or handling errors from affecting wafers or devices.

Equipment Uptime, Recipes, and Tool Matching

Term	Working meaning
tool uptime	Percentage of time equipment is available and qualified for production use.
preventive maintenance	Planned equipment service performed to reduce unplanned downtime, drift, contamination, safety risk, or tool instability.
recipe	Controlled equipment parameters used to run a process step on a wafer, lot, or tool.
tool matching	Effort to make similar manufacturing tools produce equivalent results within defined limits.

Packaging, Test, and Reliability Qualification

Term	Working meaning
package	Protective and electrical interface that connects a semiconductor die to a board or system.
binning	Sorting tested semiconductor units into performance, power, speed, or quality categories.
burn-in	Stress testing used to screen for early-life failures before product release or shipment.
qualification	Evidence-based approval that a process, product, tool, package, supplier, or change meets defined requirements.

Foundry, Tape-Out, PDK, and Capacity Communication

Term	Working meaning
foundry	Semiconductor manufacturer that fabricates chips for external customers or design companies.
tape-out	Final release of a chip design to the foundry for mask generation and fabrication.
PDK	Process design kit; foundry-provided design rules, models, and files used to design chips for a process.
capacity allocation	Decision process for assigning limited foundry, tool, test, or assembly capacity across products or customers.

Industry-Specific Meeting Moves

Situation	Useful language
Wafer Fabrication Flow and Process Integration	Before we commit, I want to confirm wafer, fab, the owner, and the evidence behind the decision. If process flow, route control, layer dependency, and integration risk must be confirmed before movement., I recommend we document the risk and agree on the next step.
Lithography, Reticles, and Critical Dimensions	Before we commit, I want to confirm lithography, photoresist, the owner, and the evidence behind the decision. If reticle status, photoresist behavior, exposure conditions, metrology repeatability, and control limits need review., I recommend we document the risk and agree on the next step.
Deposition, Etch, CMP, and Process Windows	Before we commit, I want to confirm deposition, etch, the owner, and the evidence behind the decision. If deposition uniformity, etch selectivity, cmp margin, and the qualified process window must be protected., I recommend we document the risk and agree on the next step.

Situation	Useful language
Metrology, SPC, and Yield Learning	Before we commit, I want to confirm metrology, SPC, the owner, and the evidence behind the decision. If spc trends, sampling change, tool history, defect signatures, and product mix must be separated., I recommend we document the risk and agree on the next step.
Defect Density and Cleanroom Contamination	Before we commit, I want to confirm defect density, particle, the owner, and the evidence behind the decision. If defect density, cleanroom protocol, contamination source, containment, and affected-lot traceability require action., I recommend we document the risk and agree on the next step.
Equipment Uptime, Recipes, and Tool Matching	Before we commit, I want to confirm tool uptime, preventive maintenance, the owner, and the evidence behind the decision. If tool uptime, preventive maintenance status, recipe qualification, tool matching evidence, and bottleneck risk must be balanced., I recommend we document the risk and agree on the next step.
Packaging, Test, and Reliability Qualification	Before we commit, I want to confirm package, binning, the owner, and the evidence behind the decision. If package interaction, binning criteria, burn-in results, qualification status, and customer-use conditions are not interchangeable., I recommend we document the risk and agree on the next step.
Foundry, Tape-Out, PDK, and Capacity Communication	Before we commit, I want to confirm foundry, tape-out, the owner, and the evidence behind the decision. If foundry allocation, pdk readiness, mask schedule, change freeze, and capacity allocation need documented assumptions., I recommend we document the risk and agree on the next step.

High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.
- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

Practical Collocations and Field Moves

These phrases are intentionally practical. Learners should rehearse them as complete workplace moves: what they mean, when to use them, what evidence they require, and what decision they support.

Collocation set 1

Collocation	How it works in the field
put a lot on hold	Stop a wafer lot from moving until the hold reason, owner, risk, and release condition are clear.
release a lot	Allow a held lot to continue only after route, disposition, and risk acceptance have been documented.
open an excursion	Start a formal investigation when SPC, defect, yield, or tool behavior moves outside expected limits.
contain affected lots	Identify, quarantine, or restrict lots that may share the same exposure, tool path, material, or time window.
verify lot genealogy	Trace where a lot has been, which tools touched it, and which lots share possible exposure.
quarantine suspect material	Physically or systemically prevent wafers, chemicals, parts, or finished units from being used or shipped.
review SPC charts	Look at control charts, rules, limits, and trends before deciding whether a signal is real.
separate signal from noise	Avoid reacting to a single data point until sampling, repeatability, and tool history are checked.
tighten the process window	Reduce allowed process variation when yield, reliability, or margin is at risk.

Collocation	How it works in the field
run split lots	Process comparable lots or wafers under different conditions to isolate a variable or compare paths.

Collocation set 2

Collocation	How it works in the field
qualify a recipe	Generate evidence that a controlled tool recipe meets process, quality, and reliability requirements.
match a tool	Show that two tools produce equivalent results within approved limits for a given process step.
move lots to the matched tool	Transfer work to an alternate tool only within the approved matching, recipe, and product constraints.
pull in preventive maintenance	Move planned maintenance earlier to reduce drift, contamination, safety, or downtime risk.
recover tool uptime	Restore qualified equipment availability without bypassing restart, monitor, or qualification requirements.
check chamber history	Review recent maintenance, cleans, alarms, consumables, recipes, and abnormal events for a process chamber.
confirm metrology repeatability	Repeat or cross-check measurements before treating a trend as a process problem.
hold shipment	Prevent finished units from leaving until quality, reliability, customer, or compliance questions are resolved.
screen for early-life failure	Use burn-in, stress, or test screening to identify infant mortality risk before release.
bin devices by speed and power	Sort parts into usable categories after test, often tied to customer specifications or price.

Collocation set 3

Collocation	How it works in the field
perform failure analysis	Investigate failed die, packages, or returned units to identify physical, electrical, process, or use-related causes.
close the 8D	Complete the structured customer problem-solving report with verified root cause and effective corrective action.
freeze the mask set	Stop design changes so masks, reticles, wafer starts, and downstream schedules can proceed.
release the PDK	Publish foundry design rules, models, and files for a process after readiness checks are complete.
lock the tape-out package	Finalize the files, checks, signoffs, and assumptions needed for mask generation.
allocate wafer starts	Assign constrained foundry capacity or fab starts across products, customers, or priorities.
approve a deviation	Permit a controlled exception only after impact, owner, duration, and risk acceptance are documented.
issue a customer quality notice	Give customers a factual update about risk, containment, impact, and next action without speculation.

Specialized Nomenclature Expansion

Use this expansion set for learners who already know the general module terms and need the more specific nouns, acronyms, and measurement language that appear in fab, test, packaging, reliability, and customer-quality conversations.

Fab logistics

Term	Working meaning
FOUP	Front opening unified pod used to transport and protect wafers in automated fab environments.
lot traveler	Record or system route showing where the lot goes, what has been done, and what instructions apply.
hold code	System code that explains why material is blocked from movement or shipment.
WIP	Work in process; material currently moving through manufacturing, test, assembly, or disposition.
queue time	Time a lot waits between process steps; excessive queue time can affect quality or cycle time.
lot genealogy	Traceable history of tools, materials, routes, rework, holds, and related lots.

Process control

Term	Working meaning
control wafer	Wafer used to monitor process or tool behavior without risking production material.
split lot	Lot divided or compared across conditions to test a process variable or disposition path.
golden tool	Reference tool whose performance is treated as the comparison point for matching.
chamber matching	Effort to align performance across chambers in the same or similar equipment.
inline monitor	Measurement taken during manufacturing to catch drift before final test or yield loss.
PCM	Process control monitor structures or measurements used to evaluate process health.

Lithography

Term	Working meaning
overlay	Alignment accuracy between patterned layers on the wafer.
dose	Exposure energy applied during lithography.
focus	Exposure focus condition that affects printed feature quality.
OPC	Optical proximity correction used to adjust mask patterns for printing behavior.
CD-SEM	Critical-dimension scanning electron microscope used to measure small patterned features.
scatterometry	Optical metrology method used to infer feature profiles, dimensions, or film characteristics.

Films and etch

Term	Working meaning
film thickness	Measured thickness of a deposited or grown layer.
sheet resistance	Electrical resistance measurement used to monitor thin films or implants.
selectivity	How much faster one material is removed than another during etch or CMP.
uniformity	Degree to which a process result is consistent across wafer, lot, tool, or chamber.
end-point detection	Method for determining when an etch or process step has reached the target condition.
slurry	Abrasive chemical mixture used in CMP.
post-CMP clean	Cleaning step used to remove residue, particles, or contamination after CMP.

Test and sort

Term	Working meaning
wafer sort	Electrical testing of die while still on the wafer.
die sort	Classification of die after wafer-level testing.
probe card	Test interface that contacts wafer pads during wafer sort.
parametric test	Electrical measurement of device or process parameters, often used to monitor margins.
guardband	Extra margin used to reduce risk when specification, process, or use conditions vary.

Reliability

Term	Working meaning
infant mortality	Early-life failures that appear soon after device use or stress.
FIT rate	Failures in time; reliability measure often expressed as failures per billion device hours.
electromigration	Reliability failure mechanism caused by metal movement under current stress.
TDDDB	Time-dependent dielectric breakdown; reliability mechanism affecting insulating layers.
HCI	Hot carrier injection; reliability mechanism that can degrade transistor performance.
NBTI	Negative-bias temperature instability; reliability mechanism affecting transistor threshold behavior.

Packaging

Term	Working meaning
wire bond	Electrical connection made by bonding fine wires between die and package leads or substrate.
flip chip	Package approach where die are connected face-down through bumps.
underfill	Material placed under flip-chip die to improve mechanical reliability.
bump	Conductive connection point used in advanced packages or wafer-level packaging.
TSV	Through-silicon via used to connect vertically through silicon in advanced packages.
substrate	Package platform that supports the die and routes signals to the board.

Quality and customer

Term	Working meaning
MRB	Material review board that decides disposition for nonconforming or suspect material.
deviation	Approved exception from a normal requirement, route, specification, or process condition.
rework	Additional approved processing intended to bring material back into requirement.
scrap	Disposition for material that cannot be used or recovered acceptably.
FA	Failure analysis; structured investigation of why a device, die, package, or system failed.
decapsulation	Removal of package material to inspect die, bonds, or physical evidence during FA.

Contrast Pairs

Do not confuse	Useful distinction
wafer vs node	In wafer fabrication flow and process integration, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
lithography vs critical dimension	In lithography, reticles, and critical dimensions, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
deposition vs process window	In deposition, etch, cmp, and process windows, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
metrology vs excursion	In metrology, spc, and yield learning, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
defect density vs contamination control	In defect density and cleanroom contamination, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
tool uptime vs tool matching	In equipment uptime, recipes, and tool matching, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
package vs qualification	In packaging, test, and reliability qualification, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
foundry vs capacity allocation	In foundry, tape-out, pdk, and capacity communication, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.